

Ensemble Nonlinear: **Low-cost Ensemble Performance** **with Networked Raspberry Pi's**

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Figure 1. Ensemble Nonlinear performs using headless Raspberry Pi computers connected to a single audio server.

ABSTRACT

Musical performance ensembles featuring orchestras of laptops have become firmly established as a vehicle for advancing electronic and electroacoustic composition and research practices in institutions around the world. But while the successes of many such laptop orchestras commonly utilize high-cost homogeneous technology infrastructures featuring high-end laptops, audio interfaces and highly specialized hemispherical speakers, the financial cost of creating such an ensemble is simply out of reach for many artists, ensembles and educational institutions. This paper details the creation of Ensemble Nonlinear, an electronic music performance ensemble and research platform utilizing low-cost Raspberry Pi single-board computers (SBC) as networked clients connected to a single audio server laptop and speaker array.

1. INTRODUCTION

There already exists at this time a significant body of work descended from a particular model of ensemble networked musical performance known as the “laptop orchestra”. Foundational work carried out by Perry Cook, Dan Trueman and the researchers at the Princeton Soundlab resulted in the creation of the Princeton Laptop Orchestra or PLOrk [5, 8]. Central to the Princeton or PLOrk model was a homogeneous technology infrastructure consisting of identical laptop stations featuring hemispherical loudspeakers acting as radiant positional sound sources for each performer in the ensemble [1]. The Princeton model was further extended by Ge Wang with the inception of the Stanford Laptop Orchestra or SLOrk [6, 7], replicating the same platform homogeneity as its predecessor. And while the use of dedicated high-end Macbook Pro

laptops and custom speaker designs has allowed each of these ensembles and the academic courses built around them to minimize effort in troubleshooting multiple operating systems and instead focus effort primarily on artistic and technological development, the financial cost of building and deploying such an undertaking is simply out of reach for many schools and programs.

2. TECHNOLOGY INFRASTRUCTURE

To create a low-cost pedagogical vehicle for electronic ensemble performance, in the Spring of 2017 a new performance course and ensemble “Ensemble Nonlinear” was created at Rensselaer Polytechnic Institute in Troy, New York using Raspberry Pi SBCs as the core computing and teaching platform. Making use of existing audio infrastructure such as multi-channel speaker arrays, Ensemble Nonlinear consists of a set of Raspberry Pi’s connected to a laptop running an audio server using ethernet and Open Sound Control (OSC).



Figure 2. A standard Nonlinear setup with Raspberry Pi’s, network switch and client keyboards.

Each Raspberry Pi is running a customized headless Raspbian Lite operating system. Once powered and plugged into an ethernet switch, each Pi automatically assigns itself a fixed IP address and connects to the server laptop using Open Sound

Control. In its default configuration, a key logging script launches automatically and streams all key presses from each headless Pi to the server, acting as a



Figure 3. A single multi-channel audio interface powers the entire ensemble.

light client capable of communicating commands to audio processes running on the server with no additional configuration needed. The server passes organizational commands back to each pi using the Secure Shell Protocol (SSH) having previously been authenticated with each device’s Secure Hash Algorithm (SHA) key.

2.1 Multi-channel Support

Ensemble Nonlinear eschews the familiar hemispherical multichannel speaker arrays connected to each student workstation and instead drives audio and visual servers with control data sent from each light client machine. In this manner, sound can be spatialized using ambisonics or direct speaker panning. Only a single multi-channel audio interface is used in this configuration, greatly reducing the cost while still allowing individual spatialization of each performer’s sound around the speaker array.

2.2 Programming Paradigms

Audio programming for Ensemble Nonlinear has predominantly been handled using ChucK, allowing the ensemble to easily add instances of instruments or compositional scripts for each member of the ensemble. And by using ChucK, a large percentage of the existing catalog of musical works composed for Princeton and Stanford’s laptop orchestras can be modified to receive OSC control data and performed by Ensemble Nonlinear. At the same time, student works for the ensemble have made use of any number of audio-specific programming languages (Max/MSP, Pure Data, SuperCollider) and gaming/graphics engines (Unity, Unreal Engine, Processing, OpenGL).

2.3 User Controls

As the Raspberry Pi's used in Ensemble Nonlinear run headless without attached video displays, they must boot and be controllable immediately after being powered, with no software initialization or configuration needed. A Chuck script running on each device accesses the attached HID keyboard and streams keypress data (including ASCII code, modifier key(s) pressed, human-readable character and the device's IP address as a unique identifier).

An addition for the 2019 season of Ensemble Nonlinear was a custom Pi Wiimote driver that initializes on boot and allows any Wiimote device to be paired with a device. Button presses and accelerometer data for both a single Nintendo Wiimote controller, as well as the secondary connected Nunchuck controller are all supported. In similar fashion to the Chuck keyboard logging script, the Wiimote driver streams button and device data to the same host server via OSC.



Figure 4. A set of devices branded for the 2019 season with device names from “Guardians of the Galaxy”

3. PRIOR ART

As previously mentioned, the Princeton Laptop Orchestra and the Stanford Laptop Orchestra serve as two prototypical examples of homogeneous technology platforms for ensemble laptop performance and hemispherical speakers. The use of Raspberry Pi's as instruments in ensemble performance has been explored as part of the L2ORk Linux Laptop Orchestra [2]. Numerous other laptop orchestras and performance ensembles have explored similar paradigms of music and technology-based composition for locally networked computer systems including early work by The Hub [4] and by the Carnegie Mellon Laptop Orchestra [3].

4. PERFORMANCE PRACTICE

While many facets of laptop orchestral performance practice remain unchanged when using Ensemble Nonlinear's client-server model, one fundamental difference lies in the reliance on a centralized audio server. In its default configuration Ensemble Nonlinear's Raspberry Pi 3's stream Open Sound Control Data only, passing on keystroke data from a QWERTY computer keyboard or accelerometer and button-press data from Wiimotes and Wii-nunchuck controllers. No audio is computed on the devices themselves, instead all audio is rendered by the audio server, itself a fundamentally more powerful machine.

Another fundamental difference between “traditional” laptop-orchestras and the Nonlinear model lies in the manner of sound spatialization and diffusion. A great strength of laptop orchestras following the Princeton and Stanford models is the use of hemispherical speakers situated within the ensemble itself, capable of radiating each performer's sound from their individual locations. In this configuration performers can situate their speaker right next to their own bodies, making it much easier to hear one's own sound amidst the ensemble. Using a single multi-channel speaker array - such as the typical ring of 8 speakers surrounding the audience used for Ensemble Nonlinear - the sonic output of the ensemble is significantly different. For a large ensemble, this can create a situation in which performers have significant difficulty in hearing their own sound.

That being said, Ensemble Nonlinear is a relatively new performance ensemble and course at Rensselaer, and as such class sizes to date have been relatively small (between four and twelve students), shifting the style of performance practice more towards a model of electronic chamber music rather than that of an orchestra. In this context, the ensemble performs using a non-localized speaker array situated around the edges of the performance space with performers seated near each speaker, minimizing issues for each performer's ability to differentiate their own instrumental sounds from those of other members of the ensemble.



Figure 5. Students perform electronic chamber music using a simple four-channel speaker system (top). Students in the 128-channel CRAIVE lab. (bottom)

5. REPERTOIRE

The adaptation of musical works from existing laptop orchestras has been surprisingly straightforward in the Ensemble Nonlinear client-server model. The ensemble's current repertoire combines works composed for the PLOrk and SLOrk with new works written for Ensemble Nonlinear. A sample selection of early works from the first three years of the ensemble is provided here.

5.1 *alone+easy*

This work was originally written for the Stanford Laptop Orchestra and reworked for Ensemble Nonlinear. *alone+easy* allows each performer to control a complex granulator instance on the ChuckK server, granulating pre-recorded guitar feedback and excerpts from a mix of rock songs. For a solo performance at ICMC 2015, this composition was re-written to accept OSC keyboard control and as such has served as the prototypical model for Ensemble Nonlinear's technology stack.

5.2 *Clix*

Clix, by Ge Wang, uses key presses on the computer keyboard as triggers for noise bursts through tuned filters. This was the first existing work ported to Ensemble Nonlinear's client-server model.

5.3 *Jude*

Jude is a gamut-based composition based on the chord progression from the Lennon and McCartney Beatles song *Hey Jude*. Each performer linearly yet asynchronously steps through notes of the source score, playing each one on a separate instrument instance of a physically-modeled bowed string. *Jude* has been performed by Ensemble Nonlinear with live melodic accompaniment by a number of instruments including saxophones, violins and guitars.

5.4 *BBQ*

BBQ was composed by Hana Shin as a solo performance piece and was adapted by members of Ensemble Nonlinear. Letters on the computer keyboard are mapped to trigger onomatopoeic sounding samples, turning text typed by each performer into performative sound.

5.6 *Tünnin*

Tünnin was composed as a telematic collaboration between Ensemble Nonlinear and the Music Technology Improvisation ensemble at the University of Michigan for the 2018 NowNet Arts Conference on Network Music at Stony Brook University in Long Island. *Tünnin* pairs computer performers with instrumental performers, with the former granulating samples generated by the latter.

5.6 *FibSeq*

Composed by Oleg Yakovets, *FibSeq* features an unorthodox Fibonacci-based step sequencer running as a Max for Live plugin within Ableton Live, driving digital recreations of analog synths. Each performer is placed in control of various parameters of an individual track, with global changes controlled by a conductor.

6. DISCUSSION

While the initial driving force for building Ensemble Nonlinear's technology infrastructure using low-cost Raspberry Pi computers was financial, the small footprint and flexible nature of the devices have quickly revealed interesting benefits to the choice. As the entire Raspberry Pi operating system as well as all data for the device is stored on microSD flash memory cards, a configured device can be imaged, stored online, and can be copied onto a new card to initialize and bring a new device online quickly. And while the ensemble currently uses 32GB memory cards, much smaller cards

could be used, significantly reducing the time needed to flash and bring up a new machine.

As a pedagogical tool, students in the Ensemble are able to explore working in Linux and with Open Source tools and programming languages without affecting their primary computer systems. And the low cost of each device has made it easy for students to purchase their own Raspberry Pi's to work on outside of the course.

As the ensemble's title suggests, *Ensemble Nonlinear* doesn't try to restrict students to use the Raspberry Pi infrastructure for their own research and compositional directions. Instead, the platform exists as a low-cost, easy to maintain launching pad for each season's course and ensemble. As the majority of students taking the course have generally neither performed in any electronic music ensemble before nor performed in *Ensemble Nonlinear* before, the ability for the platform to act as a plug-and-play system to quickly bootstrap ensemble members into the experience has been extremely successful. Rehearsals for the course begin on day one every term, introducing students to the core repertoire mentioned previously and sparking their own creative and research directions for the coding and musical projects required in the course.

The design of Ensemble Nonlinear can scale relatively easily using existing infrastructure from the University's music and audio engineering programs, but it should be noted that this model is reliant on the availability of basic computer and audio infrastructure such as a laptop and a set of speakers. For these purposes, a set of eight powered Genelec monitors have been used, driven by a single multi-channel MOTU Ultralite Mk4 interface connected to an Apple MacBook Pro laptop with 32 GB of RAM.

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